

Chapter 2

Continuous time Markov chains

2.1 Introduction — Q-Matrices

As before we assume that we have a finite or countably infinite statespace I , but now the Markov chains $X = \{X(t) : t \geq 0\}$ have a continuous time parameter $t \in [0, \infty)$.

Definition 2.1. A collection of random variables $(X_t)_{t \geq 0}$ taking values in I has the **Markov property**, if for all $0 = t_0 < t_1 < \dots < t_{n+1}$ and all $i_0, \dots, i_{n+1} \in I$ we have

$$\mathbb{P}[X_{t_{n+1}} = i_{n+1} \mid X_{t_n} = i_n, \dots, X_0 = i_0] = \mathbb{P}[X_{t_{n+1}} = i_{n+1} \mid X_{t_n} = i_n]. \quad (*)$$

We assume **time-homogeneity**, so that there exist matrices $(P(t) : t \geq 0)$ such that the right hand side of $(*)$ equals $P(t_{n+1} - t_n)_{i_n i_{n+1}} =: p_{i_n i_{n+1}}(t_{n+1} - t_n)$. Then for an initial distribution w we have

$$\mathbb{P}_w[X_{t_1} = i_1, \dots, X_{t_n} = i_n] = \sum_{i_0 \in I} w_{i_0} p_{i_0 i_1}(t_1) p_{i_1 i_2}(t_2 - t_1) \dots p_{i_{n-1} i_n}(t_n - t_{n-1}).$$

Remark 2.2. A complete definition and construction of continuous-time Markov chains is more technical; see Theorems 2.8.2 and 2.8.4 in Norris: Markov Chains, and the unit MA30125. We do not deal with all of the technicalities in these notes.

2.1.1 Definition of a Q-matrix

Question. How to specify $(P(t) : t \geq 0)$? In discrete time we had transitions described by the matrices $I, P, P^2, \dots, P^n, \dots$. A problem we are immediately faced with is that in continuous time there is no smallest positive time, so we cannot use the powers of a transition matrix. The correct analogy is an exponential function $P(t) = e^{tQ}$, $t \geq 0$, for some matrix Q . When the state space I is finite, the exponential of a matrix can indeed be defined without difficulty.

Definition 2.3. For a finite square matrix A , we define $e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$.

The series converges absolutely elementwise (see Problem Sheets).

When I is infinite, the series may not converge. However, throughout this chapter, we will be able to use the finite case as a guide as to what the correct definitions might be for an infinite state space. The first of these is that Q needs to satisfy some requirements.

Question. What kind of matrix Q should be?

Definition 2.4. Let I be a countable set. A **Q-matrix** on I is a matrix $Q = (q_{ij} : i, j \in I)$ such that

- (i) $0 \leq q_i := -q_{ii} < \infty$ for all $i \in I$;
- (ii) $q_{ij} \geq 0$ for all $i, j \in I, i \neq j$;
- (iii) $\sum_{j \in I} q_{ij} = 0$ for all $i \in I$ (row sums of Q are 0).

Question. What is the probabilistic meaning of Q ?

We first state the answer to this question, then discuss it in a number of examples, before explaining the analytical reasons.

When in state i , the Markov chain will stay in i for an $\text{Exp}(q_i)$ distributed time.

We need to slightly extend the notion of exponential random variable to deal with situations where $q_i = 0$. This is given in the next definition.

Definition 2.5. Let $q \in [0, \infty)$. We say that a random variable T is $\text{Exp}(q)$, shortly written $T \sim \text{Exp}(q)$, if for all $t \geq 0$ we have $\mathbb{P}[T > t] = e^{-qt}$. Note that we do allow $q = 0$, in which case $T \equiv \infty$.

We call q_i the rate of leaving state i .

When the Markov chain jumps, it will jump from state i to state $j \neq i$ with probability q_{ij}/q_i , independently of the length of stay in i and previous movements. Note: $\sum_{j:j \neq i} q_{ij} = -q_{ii} = q_i$, thus the given probabilities indeed add up to 1.

We call the q_{ij} ($j \neq i$) the **the rate of jump from i to j** .

2.1.2 Examples

We next give some examples of Q-matrices.

Example 2.6.

A bird makes visits from its nest to a meadow at exponential rate 1 and to a lake at exponential rate 2. When at the meadow, the bird looks for food there for a time that is $\text{Exp}(4)$ distributed. When finished, it returns to its nest with probability $3/4$, and flies to the lake with probability $1/4$. When at the lake, the bird looks for food there for an $\text{Exp}(8)$ time. When finished, it returns to its nest with probability $3/4$, and flies to the meadow with probability $1/4$.

Let $I = \{N, M, L\}$ denote the state space. From the text, we have $q_M = 4$ and $q_{MN}/q_M = 3/4$, $q_{NL}/q_M = 1/4$. This gives the entries $q_{MM} = -4$, $q_{MN} = 3$, $q_{ML} = 1$. Similarly, we have $q_L = 8$ and $q_{LN}/q_L = 3/4$, $q_{LM}/q_L = 1/4$, which gives the entries $q_{LL} = -8$, $q_{LN} = 6$, $q_{LM} = 2$. Finally, we have $q_{NM} = 1$ and $q_{NL} = 2$, which give $q_N = 3$ and $q_{NN} = -3$. Thus the Q-matrix is

$$Q = \begin{pmatrix} -3 & 1 & 2 \\ 3 & -4 & 1 \\ 6 & 2 & -8 \end{pmatrix}.$$

As the following lemma shows, when the bird is at its nest, we can think of having two independent ‘exponential clocks’ with rates $q_{NM} = 1$ and $q_{NL} = 2$. When one of these clocks rings, the bird moves to that place.

Lemma 2.7. *Let $T_1 \sim \text{Exp}(\lambda_1)$ and $T_2 \sim \text{Exp}(\lambda_2)$ be independent. Then we have:*

- (i) $T := \min\{T_1, T_2\} \sim \text{Exp}(\lambda_1 + \lambda_2)$;
- (ii) $\mathbb{P}[T = T_1] = \lambda_1/(\lambda_1 + \lambda_2)$ and $\mathbb{P}[T = T_2] = \lambda_2/(\lambda_1 + \lambda_2)$;
- (iii) The events $\{T = T_k\}$, $k = 1, 2$ are independent of T .

Proof. First note: $\mathbb{P}[T_1 = T_2] = 0$, since T_1 and T_2 are independent continuous random variables. We compute:

$$\begin{aligned}
 \mathbb{P}[T = T_1, T > t] &= \mathbb{P}[T_1 > t, T_2 > T_1] = \int_t^\infty f_{T_1}(s) \mathbb{P}[T_2 > s \mid T_1 = s] ds \\
 &= \int_t^\infty f_{T_1}(s) \mathbb{P}[T_2 > s] ds = \int_t^\infty \lambda_1 e^{-\lambda_1 s} e^{-\lambda_2 s} ds = \int_t^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)s} ds \quad (2.1) \\
 &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \int_t^\infty (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)s} ds = \frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t}.
 \end{aligned}$$

Analogously we obtain:

$$\mathbb{P}[T = T_2, T > t] = \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t}. \quad (2.2)$$

Setting $t = 0$ in these formulas gives $\mathbb{P}[T = T_1] = \lambda_1/(\lambda_1 + \lambda_2)$ and $\mathbb{P}[T = T_2] = \lambda_2/(\lambda_1 + \lambda_2)$. Adding the two formulas gives:

$$\mathbb{P}[T > t] = e^{-(\lambda_1 + \lambda_2)t}.$$

The independence claim (iii) follows from (2.1) and (2.2). $\square$

Example 2.8. (Poisson process)

Let $\lambda > 0$ be parameter. A **Poisson process of rate λ** is a random process that visits the states $0, 1, 2, \dots$ in sequence with independent $\text{Exp}(\lambda)$ times between jumps. The diagram of the process is:



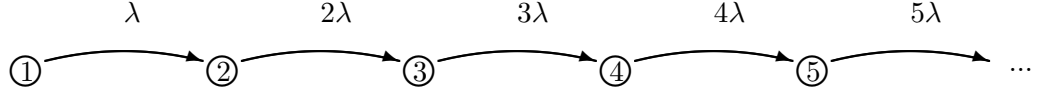
The Q-matrix is given by $q_{i,i+1} = \lambda$, $i \geq 0$ and $q_{i,i} = -\lambda$, $i \geq 0$, with all other entries 0:

$$Q = \begin{pmatrix} -\lambda & \lambda & & & \\ & -\lambda & \lambda & & \\ & & \ddots & \ddots & \\ & & & \ddots & \ddots \end{pmatrix}.$$

It can be derived (non-examinable) that $\mathbb{P}_0[X_t = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$, so that starting in state 0 we have $X_t \sim \text{Poisson}(\lambda t)$.

Example 2.9. (Pure birth process)

Suppose that each individual in a population (of bacteria, say) gives birth to a new individual at rate λ , independently of other individuals. If there are i individuals at present, let $T_1, \dots, T_i \sim \text{Exp}(\lambda)$ be the times they will next give birth. A new individual is added to the population at time $\min\{T_1, \dots, T_i\} \sim \text{Exp}(i\lambda)$. (This statement uses a generalization of Lemma 2.7 to more than two exponentials, and can be derived similarly.) Thus the process has diagram:



The process visits the states $1, 2, 3, \dots$ in sequence. The times between jumps are independent random variables, with the jump from i to $i + 1$ distributed as $\text{Exp}(i\lambda)$. The Q-matrix is given by $q_{ii+1} = i\lambda$, $i \geq 1$ and $q_{ii} = -i\lambda$, $i \geq 1$, with all other entries 0:

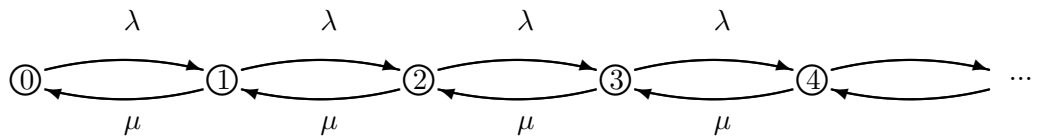
$$Q = \begin{pmatrix} -\lambda & \lambda & & & \\ & -2\lambda & 2\lambda & & \\ & & -3\lambda & 3\lambda & \\ & & & \ddots & \ddots \\ & & & & \ddots & \ddots \end{pmatrix}.$$

Example 2.10. (The M/M/1 queue)

Customers arrive to a queue according to a Poisson process of rate λ . There is a single server, and service times are independent $\text{Exp}(\mu)$ random variables. Let

$$X_t = \# \text{ customers in the queue at time } t,$$

(including any customer being served). The rates can be derived as follows. If the queue is empty, a new customer arrives after an $\text{Exp}(\lambda)$ time, so $q_{01} = \lambda$. If the queue is non-empty, the length increases by 1 at rate λ and decreases by 1 at rate μ , determined by independent $\text{Exp}(\lambda)$ and $\text{Exp}(\mu)$ random variables that give the arrival time of the next customer and the service completion of the currently served customer. Whether jump is up or down is determined by which variable is smaller. The diagram of the process is:



The Q-matrix is:

$$Q = \begin{pmatrix} -\lambda & \lambda & & & \\ \mu & -(\lambda + \mu) & \lambda & & \\ & \mu & -(\lambda + \mu) & \lambda & \\ & & \ddots & \ddots & \ddots \end{pmatrix}.$$

Question: If there are currently two customers in the queue, how long will it take before the queue length changes? Since $q_2 = -q_{22} = \lambda + \mu$, the answer is that it takes an $\text{Exp}(\lambda + \mu)$ distributed time.

Question: With what probability will the queue length increase to three? The answer is given by $q_{23}/q_2 = \lambda/(\lambda + \mu)$.

2.1.3 Infinitesimal interpretation of the Q-matrix

We now make some computations in the case of finite I to see that Q gives probabilities on (infinitesimally) short time intervals. Since

$$P(t) = I + tQ + O(t^2),$$

we have

$$p_{ii}(h) = I_{ii} + (hQ)_{ii} + o(h) = 1 + q_{ii}h + o(h) = 1 - q_i h + o(h), \quad \text{as } h \downarrow 0.$$

(Recall that $o(h)$ denotes a term that is smaller order than h , i.e. $o(h)/h \rightarrow 0$ as $h \downarrow 0$.) Thus q_i , the rate of leaving state i , gives the asymptotic probability that we leave state i in a short time interval.

Similarly,

$$p_{ij}(h) = I_{ij} + (hQ)_{ij} + o(h) = q_{ij}h + o(h), \quad \text{as } h \downarrow 0.$$

Thus q_{ij} , the rate of jump from state i to state j gives the asymptotic probability that the chain moves from i to j in a short time interval.

In the M/M/1 queue example, a new customer will arrive in a short time interval of duration h , with probability $\lambda h + o(h)$. If the queue is non-empty, a customer will finish service in a short time interval of duration h , with probability $\mu h + o(h)$. In the same time interval, no jump will occur with probability $1 - (\lambda + \mu)h + o(h)$.

2.2 Kolmogorov's equations

The Q-matrix provides a system of differential equations that $P(t)$ satisfies.

Proposition 2.11. *Let Q be a Q-matrix on a finite state space I , and $P(t) = e^{tQ}$, $t \geq 0$. Then*

- (i) $\frac{d}{dt}P(t) = P(t)Q$, $P(0) = I$ (Kolmogorov's forward equation);
- (ii) $\frac{d}{dt}P(t) = QP(t)$, $P(0) = I$ (Kolmogorov's backward equation);
- (iii) $P(t+s) = P(t)P(s)$ for all $s, t \geq 0$. In coordinate form:

$$p_{ij}(t+s) = \sum_{k \in I} p_{ik}(t) p_{kj}(s), \quad i, j \in I, \quad (\text{Chapman-Kolmogorov equations}).$$

Proof. (Non-examinable) Statements (i) and (ii) follow by differentiating the power series for e^{tQ} term-by-term:

$$\frac{d}{dt} \left[\sum_{n=0}^{\infty} \frac{t^n Q^n}{n!} \right] = \sum_{n=1}^{\infty} \frac{n t^{n-1} Q^n}{n(n-1)!} = \sum_{n=1}^{\infty} \frac{t^{n-1} Q^n}{(n-1)!} = P(t)Q = QP(t).$$

Statement (iii) follows from $e^{(t+s)Q} = e^{tQ} e^{sQ}$, which can be proved by expanding $(tQ+sQ)^n$ according to the Binomial theorem, as for the scalar exponential function. $\square$

The probabilistic meaning of part (iii) is that:

$$\begin{aligned} p_{ij}(t+s) &= \mathbb{P}_i[X_{t+s} = j] = \sum_{k \in I} \mathbb{P}_i[X_t = k, X_{t+s} = j] = \sum_{k \in I} \mathbb{P}_i[X_t = k] \mathbb{P}_i[X_{t+s} = j \mid X_t = k] \\ &= \sum_{k \in I} \mathbb{P}_i[X_t = k] \mathbb{P}_k[X_s = j] = \sum_{k \in I} p_{ik}(t) p_{kj}(s). \end{aligned}$$

2.3 Resolvents

Let Q be the Q -matrix of the Markov chain X and $P(t), t \geq 0$ the transition matrix function.

2.3.1 Laplace transforms

A function $f : [0, \infty) \rightarrow \mathbf{R}$ is called *good* if f is continuous and either bounded (there exists $K > 0$ such that $|f(t)| \leq K$ for all t) or integrable ($\int_0^\infty |f(t)| dt < \infty$) or both. If f is good we can associate the Laplace transform $\hat{f} : (0, \infty) \rightarrow \mathbf{R}$ which is defined by

$$\hat{f}(\lambda) = \int_0^\infty e^{-\lambda t} f(t) dt.$$

An important **example** is the case $f(t) = e^{-\alpha t}$. Then

$$\hat{f}(\lambda) = \int_0^\infty e^{-\lambda t} e^{-\alpha t} dt = \frac{1}{\lambda + \alpha}.$$

An important result about Laplace transforms is the following:

Theorem 2.12 (Uniqueness Theorem). *If f and g are good functions on $[0, \infty)$ and $\hat{f}(\lambda) = \hat{g}(\lambda)$ for all $\lambda > 0$. Then $f = g$.*

This implies that we can *invert* Laplace transforms uniquely, at least when restricting attention to good functions.

Example 2.13. If $\hat{f}(\lambda) = \frac{2}{5} \frac{1}{\lambda+3}$, then $f(t) = \frac{2}{5} e^{-3t}$.

2.3.2 Resolvents

The basic idea here is to calculate the exponential $P(t) = e^{Qt}$ of the Q -matrix using the Laplace transform.

Definition 2.14. The matrix function

$$R(\lambda) := \int_0^\infty e^{-\lambda t} P(t) dt, \quad \lambda > 0,$$

is called the **resolvent** of Q . (The integration of the matrix function is understood componentwise.)

Lemma 2.15. *Let Q be a Q -matrix on a finite state space I .*

(i) *For all $\lambda > 0$ we have $e^{-t(\lambda I - Q)} \rightarrow 0$, as $t \rightarrow \infty$.*

(ii) *For all $\lambda > 0$ we have $R(\lambda) = (\lambda I - Q)^{-1}$.*

Proof. (i) (Non-examinable) This can be proved using that Q has no eigenvalues with a positive real part.

(ii) We have

$$R(\lambda) = \int_0^\infty \left(e^{-\lambda t} I \right) e^{tQ} dt = \int_0^\infty e^{-t\lambda I} e^{tQ} dt = \int_0^\infty e^{-t(\lambda I - Q)} dt.$$

Now for any square matrix A we have

$$A \int_0^T e^{-tA} dt = \int_0^T A e^{-tA} dt = [-e^{-tA}]_{t=0}^{t=T} = -e^{-TA} + I.$$

Thus if $e^{-TA} \rightarrow 0$ as $T \rightarrow \infty$, we have $\int_0^\infty e^{-tA} dt = A^{-1}$. Using this with $A = (\lambda I - Q)$ we get $R(\lambda) = (\lambda I - Q)^{-1}$. $\square$

Now $R(\lambda) = (\lambda I - Q)^{-1}$, if it exists, can be calculated and $P(t)$ can be recovered by inverting the Laplace transform. The components of $R(\lambda)$ are the Laplace transforms of the functions p_{ij} ,

$$r_{ij}(\lambda) := \int_0^\infty e^{-\lambda t} p_{ij}(t) dt = \hat{p}_{ij}(\lambda).$$

We now discuss the use of resolvents in calculations for continuous-time Markov chains using the following problem:

Problem: Consider a continuous-time Markov chain X with state space $I := \{A, B, C\}$ and transitions between the states described as follows:

- When currently in state A at time t , in the next time interval $(t, t+h)$ of *small* length h and independent of the past behaviour of the chain, with
 - probability $h + o(h)$ the chain jumps into state B ,
 - probability $2h + o(h)$ the chain jumps into state C ,
 - probability $1 - 3h + o(h)$ the chain remains in state A .
- Whilst in state B , the chain tries to enter state C at rate 2. The chain cannot jump into state A directly from state B .
- On entering state C , the chain remains there for an independent exponential amount of time of rate 3 before jumping. When the jump occurs, with probability $2/3$ it is into state A , and with probability $1/3$ the jump is to state B .

Question (a). Give the Q -matrix of the continuous-time Markov chain X .

For the jumps from state C we are given $3 = q_{CA} + q_{CB}$, the rate of leaving state C . The probability that the first jump goes to state A is $q_{CA}/(q_{CA} + q_{CB})$, which is given as $2/3$. Hence we get $q_{CA} = 2$ and $q_{CB} = 1$. Altogether,

$$Q = \begin{pmatrix} -3 & 1 & 2 \\ 0 & -2 & 2 \\ 2 & 1 & -3 \end{pmatrix}.$$

Question (b). If the Markov chain is initially in state A , that is $X_0 = A$, what is the probability that the Markov chain is in state A at time t , that is, $\mathbb{P}_A\{X(t) = A\}$?

We now solve (b) with the resolvent method. Recall

$$p_{AA}(t) = \mathbb{P}_A\{X(t) = A\}, \text{ and } r_{AA}(\lambda) = \int_0^\infty e^{-\lambda t} p_{AA}(t) dt = \hat{p}_{AA}(\lambda).$$

As $R(\lambda) = (\lambda I - Q)^{-1}$ we start by inverting

$$\lambda I - Q = \begin{pmatrix} \lambda + 3 & -1 & -2 \\ 0 & \lambda + 2 & -2 \\ -2 & -1 & \lambda + 3 \end{pmatrix}.$$

First,

$$\begin{aligned} \det(\lambda I - Q) &= (\lambda + 2) \det \begin{pmatrix} \lambda + 3 & -2 \\ -2 & \lambda + 3 \end{pmatrix} + 2 \det \begin{pmatrix} \lambda + 3 & -1 \\ -2 & -1 \end{pmatrix} \\ &= \lambda(\lambda + 3)(\lambda + 5). \end{aligned}$$

(Side remark: λ is always a factor in this determinant, because Q is a singular matrix. Recall that the sum over the rows is zero!) Now the inverse of a matrix A is given by

$$(A^{-1})_{ij} = \frac{(-1)^{i+j}}{\det A} \det(M_{ji}),$$

where M_{ji} is the matrix with row j and column i removed. Hence, in our example, the entry in position AA is obtained by

$$r_{AA}(\lambda) = (\lambda I - Q)_{AA}^{-1} = \frac{\det \begin{pmatrix} \lambda + 2 & -2 \\ -1 & \lambda + 3 \end{pmatrix}}{\det(\lambda I - Q)} = \frac{\lambda^2 + 5\lambda + 4}{\lambda(\lambda + 3)(\lambda + 5)}.$$

It remains to invert the Laplace transform. For this purpose we need to form partial fractions. Solving

$$\lambda^2 + 5\lambda + 4 = \alpha(\lambda + 3)(\lambda + 5) + \beta\lambda(\lambda + 5) + \gamma\lambda(\lambda + 3)$$

gives $\alpha = 4/15$ (plug $\lambda = 0$) and $\beta = 1/3$ (plug $\lambda = -3$) and $\gamma = 2/5$ (plug $\lambda = -5$). Hence

$$r_{AA}(\lambda) = \frac{4}{15\lambda} + \frac{1}{3(\lambda + 3)} + \frac{2}{5(\lambda + 5)}.$$

Now the inverse Laplace transform of $1/(\lambda + \beta)$ is $e^{-\beta t}$. We thus get

$$p_{AA}(t) = \frac{4}{15} + \frac{1}{3}e^{-3t} + \frac{2}{5}e^{-5t},$$

and note that $p_{AA}(0) = 1$ as it should be!

Question (c). Starting from state C at time 0, show that the probability X is in state B at time t is given by

$$\frac{1}{3} - \frac{1}{3}e^{-3t} \text{ for } t \geq 0.$$

Now solve part (c). To find $\mathbb{P}_C\{X_t = B\}$ consider

$$r_{CB}(\lambda) = \int_0^\infty e^{-\lambda t} p_{CB}(t) dt = \hat{p}_{CB}(t).$$

We need to find the CB coefficient of the matrix $(\lambda I - Q)^{-1}$, recalling the inversion formula we get

$$r_{CB}(\lambda) = \frac{-\det \begin{pmatrix} \lambda+3 & -1 \\ -2 & -1 \end{pmatrix}}{\lambda(\lambda+3)(\lambda+5)} = \frac{\lambda+5}{\lambda(\lambda+3)(\lambda+5)} = \frac{1}{\lambda(\lambda+3)} = \frac{1}{3} \left(\frac{1}{\lambda} - \frac{1}{\lambda+3} \right),$$

and inverting Laplace transforms gives

$$\mathbb{P}_C\{X_t = B\} = p_{CB}(t) = \frac{1}{3} - \frac{1}{3}e^{-3t}, \quad t \geq 0.$$

Resolvents have a probabilistic interpretation. Suppose we have an alarm-clock which rings, independently of the Markov chain, at a random time T , which is exponentially distributed with rate λ , i.e.

$$\mathbb{P}\{T > t\} = e^{-\lambda t}, \quad f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

What is the state of the Markov chain when the alarm clock rings? The probability of being in state j is

$$\mathbb{P}_i\{X_T = j\} = \int_0^\infty f_T(t) \mathbb{P}_i\{X_t = j \mid T = t\} dt = \int_0^\infty f_T(t) \mathbb{P}_i\{X_t = j\} dt = \int_0^\infty \lambda e^{-\lambda t} p_{ij}(t) dt = \lambda r_{ij}(\lambda),$$

hence

$$\boxed{\mathbb{P}_i\{X_T = j\} = \lambda r_{ij}(\lambda) \text{ for } T \sim \text{Exp}(\lambda), \text{ independent of } X.}$$

What is the expected time spent in state j before the alarm clock rings? This is

$$\begin{aligned} \mathbf{E}_i[\text{time spent in state } j \text{ before } T] &= \mathbf{E}_i \left[\int_0^T \mathbf{1}_{X_t=j} dt \right] = \mathbf{E}_i \left[\int_0^\infty \mathbf{1}_{X_t=j} \mathbf{1}_{t < T} dt \right] \\ &= \int_0^\infty \mathbf{E}_i [\mathbf{1}_{X_t=j} \mathbf{1}_{t < T}] dt = \int_0^\infty \mathbb{P}_i[X_t = j] \mathbb{P}[T > t] dt \\ &= \int_0^\infty e^{-\lambda t} p_{ij}(t) dt = r_{ij}(\lambda). \end{aligned}$$

In the third step, we interchanged an integral and an expectation. We will not justify this step here, only note that this is possible due to a general theorem, called Fubini's Theorem (see the unit MA40042 Measure Theory on how it is proved).

Question (d). Initially, X is in state C . Find the distribution for the position of X after an independent exponential time, T , of rate 4 has elapsed. In particular, you should find

$$\mathbb{P}_C\{X_T = B\} = \frac{1}{7}.$$

Suppose for part (d) that T is exponentially distributed with rate λ and independent of the chain. Recall,

$$\mathbb{P}_i\{X_T = j\} = \lambda r_{ij}(\lambda),$$

and by matrix inversion,

$$r_{CA}(\lambda) = \frac{\det \begin{pmatrix} 0 & \lambda+2 \\ -2 & -1 \end{pmatrix}}{\det(\lambda I - Q)} = \frac{2(\lambda+2)}{\lambda(\lambda+3)(\lambda+5)}.$$

As T is exponential with parameter $\lambda = 4$ we get (no Laplace inversion needed!)

$$\mathbb{P}_C\{X_T = A\} = \frac{2(4+2)}{(4+3)(4+5)} = \frac{12}{63} = \frac{4}{21}.$$

Similarly,

$$r_{CB}(\lambda) = \frac{-\det \begin{pmatrix} \lambda+3 & -1 \\ -2 & -1 \end{pmatrix}}{\det(\lambda I - Q)} = \frac{\lambda+5}{\lambda(\lambda+3)(\lambda+5)},$$

and

$$r_{CC}(\lambda) = \frac{\det \begin{pmatrix} \lambda+3 & -1 \\ 0 & \lambda+2 \end{pmatrix}}{\det(\lambda I - Q)} = \frac{(\lambda+3)(\lambda+2)}{\lambda(\lambda+3)(\lambda+5)}.$$

Plugging $\lambda = 4$ we get

$$\mathbb{P}_C\{X_T = B\} = 4r_{CB}(4) = \frac{1}{7} \text{ and } \mathbb{P}_C\{X_T = C\} = \frac{4+2}{4+5} = \frac{2}{3}.$$

At this point it is good to check that the three values add up to one and thus define a probability distribution on the statespace $I = \{A, B, C\}$.

2.3.3 First hitting times

Let $T_j = \inf\{t > 0 : X_t = j\}$ be the first positive time we are in state j . Define, for $i, j \in I$,

$$F_{ij}(t) = \mathbb{P}_i\{T_j \leq t\} = \mathbb{P}\{T_j \leq t \mid X(0) = i\},$$

then $F_{ij}(t)$ is the probability that we have entered state j at some time prior to time t , if we start the chain in i . We have the *first hitting time matrix function*,

$$F(t) = \left(F_{ij}(t) : i, j \in I, t \geq 0 \right),$$

with $F(0) = I$ and

$$F_{ij}(t) = \int_0^t f_{ij}(s) ds = \int_0^t \mathbb{P}_i\{T_j \in ds\},$$

where $f_{ij}(t)$ is the first hitting time density. The matrix functions $f = (f_{ij}(t), t \geq 0)$ and P are linked by the integral equation

$$\boxed{p_{ij}(t) = \int_0^t f_{ij}(s)p_{jj}(t-s) ds, \text{ for } i, j \in I \text{ and } t \geq 0.} \quad (2.3)$$

This can be checked as follows, use the strong Markov property, which is the fact that the chain starts afresh at time T_j in state j and evolves independently of everything that happened before time T_j to get

$$\begin{aligned} p_{ij}(t) &= \mathbb{P}_i\{X_t = j\} = \mathbb{P}_i\{X_t = j; T_j \leq t\} = \int_0^t \mathbb{P}_i\{X_t = j, T_j \in ds\} \\ &= \int_0^t \mathbb{P}_i\{T_j \in ds\} \mathbb{P}_j\{X_{t-s} = j\} = \int_0^t f_{ij}(s)p_{jj}(t-s) ds. \end{aligned}$$

Given two functions $f, g : [0, \infty) \rightarrow \mathbf{R}$ we can define the *convolution function* $f * g$ by

$$f * g(t) = \int_0^t f(s)g(t-s) ds,$$

and check by substituting $u = t - s$ that $f * g = g * f$. We can rewrite the integral equation (2.3) as

$$p_{ij}(t) = f_{ij} * p_{jj}(t).$$

In order to solve this equation for the unknown f_{ij} we use Laplace transforms again.

Theorem 2.16 (Convolution Theorem). *If f, g are good functions, then $\widehat{f * g}(\lambda) = \hat{f}(\lambda)\hat{g}(\lambda)$.*

To make this plausible look at the **example** of two functions $f(t) = e^{-\alpha t}$ and $g(t) = e^{-\beta t}$. Then

$$f * g(t) = \int_0^t f(s)g(t-s) ds = e^{-\beta t} \int_0^t e^{-(\alpha-\beta)s} ds = \frac{e^{-\beta t} - e^{-\alpha t}}{\alpha - \beta}.$$

Then

$$\begin{aligned} \widehat{f * g}(\lambda) &= \int_0^\infty e^{-\lambda s} (f * g)(s) ds = \frac{1}{\alpha - \beta} \left\{ \frac{1}{\lambda + \beta} - \frac{1}{\lambda + \alpha} \right\} \\ &= \frac{1}{\lambda + \alpha} \frac{1}{\lambda + \beta} = \hat{f}(\lambda)\hat{g}(\lambda). \end{aligned}$$

The convolution theorem applied to the integral equation (2.3) gives a formula for the Laplace transform of the first hitting time density

$$\boxed{\hat{f}_{ij}(\lambda) = \frac{r_{ij}(\lambda)}{r_{jj}(\lambda)} \text{ for } i, j \in I \text{ and } \lambda > 0.} \quad (2.4)$$

We use this now to answer further questions for the Problem stated in Section 2.3.2.

Question (e). Given that X starts in state C , find the probability that X enters state A at some point before time t .

We are looking for

$$\mathbb{P}_C\{T_A \leq t\} = F_{CA}(t) = \int_0^t f_{CA}(s) ds.$$

We first find the Laplace transform $\hat{f}_{CA}$ of f_{CA} , using r_{CA} and r_{AA} ,

$$\hat{f}_{CA} = \frac{\det \begin{pmatrix} 0 & \lambda + 2 \\ -2 & -1 \end{pmatrix}}{\det \begin{pmatrix} \lambda + 2 & -2 \\ -1 & \lambda + 3 \end{pmatrix}} = \frac{2(\lambda + 2)}{\lambda^2 + 5\lambda + 4},$$

and by partial fractions

$$\hat{f}_{CA} = \frac{2}{3(\lambda + 1)} + \frac{4}{3(\lambda + 4)}.$$

In this form, the Laplace transform can be inverted, which gives

$$f_{CA}(t) = \frac{2}{3}e^{-t} + \frac{4}{3}e^{-4t} \text{ for } t \geq 0.$$

Now, by integration, we get

$$\begin{aligned} \mathbb{P}_C\{T_A \leq t\} &= \int_0^t f_{CA}(s) ds \\ &= \frac{2}{3} \int_0^t e^{-s} ds + \frac{4}{3} \int_0^t e^{-4s} ds \\ &= 1 - \frac{2}{3}e^{-t} - \frac{1}{3}e^{-4t}. \end{aligned}$$

Check that $\mathbb{P}_C\{T_A \leq 0\} = 0$, as expected.

As in the case of $r_{ij} = \hat{p}_{ij}$ the Laplace transforms $\hat{f}_{ij}$ also admit a direct probabilistic interpretation with the help of an alarm clock, which goes off at a random time A , which is independent of the Markov chain X and exponentially distributed with rate λ . Indeed, recalling that $\lambda e^{-x\lambda}, x > 0$ is the density of A and f_{ij} is the density of T_j ,

$$\begin{aligned} \mathbb{P}_i\{T_j \leq A\} &= \int_0^\infty \mathbb{P}_i\{T_j \leq a\} \lambda e^{-a\lambda} da \\ &= \int_0^\infty \int_0^a f_{ij}(s) ds \lambda e^{-a\lambda} da \\ &= \int_0^\infty f_{ij}(s) \int_s^\infty \lambda e^{-a\lambda} da ds \\ &= \int_0^\infty f_{ij}(s) e^{-s\lambda} ds \\ &= \hat{f}_{ij}(\lambda). \end{aligned}$$

Altogether,

$$\boxed{\hat{f}_{ij}(\lambda) = \mathbb{P}_i\{T_j < A\} \text{ for } i, j \in I \text{ and } \lambda > 0.}$$

We use this now to answer a further question for our problem.

Question (f). What is the probability that we have reached state C at some point prior to time U , where U is an independent exponential random variable of rate $\mu = 2.5$, given that the chain is in state B at time zero?

The question asks for $\mathbb{P}_B\{T_C < U\}$ where U is exponential with rate $\mu = 2.5$ and independent of the Markov chain. We now know that

$$\mathbb{P}_B\{T_C < U\} = \hat{f}_{BC}(\mu) = \frac{r_{BC}(\mu)}{r_{CC}(\mu)} = \frac{-\det \begin{pmatrix} \mu+3 & -2 \\ 0 & -2 \end{pmatrix}}{\det \begin{pmatrix} \mu+3 & -1 \\ 0 & \mu+2 \end{pmatrix}} = \frac{2(\mu+3)}{(\mu+3)(\mu+2)} = \frac{2}{\mu+2},$$

and plugging in $\mu = 2.5$ gives $\mathbb{P}_B\{T_C < U\} = \frac{4}{9}$.

Example 2.17. A machine can be in three states A, B, C and the transition between the states is given by a continuous-time Markov chain with Q -matrix

$$Q = \begin{pmatrix} -3 & 1 & 2 \\ 3 & -3 & 0 \\ 1 & 2 & -3 \end{pmatrix}.$$

At time zero the machine is in state A . Suppose a supervisor arrives for inspection at the site of the machine at an independent exponential random time T , with rate $\lambda = 2$. What is the probability that he has arrived when the machine is in state C for the first time?

We write $T_C = \inf\{t > 0 : X_t = C\}$ for the first entry and $S_C = \inf\{t > T_C : X_t \neq C\}$ for the first exit time from C . Then the required probability is $\mathbb{P}_A\{T_C < T, S_C > T\}$. Note that $J = S_C - T_C$ is the length of time spent by the chain in state C during the first visit. This time is exponentially distributed with rate $q_{CA} + q_{CB} = 3$ and we get

$$\begin{aligned} \mathbb{P}_A\{T_C < T, S_C > T\} &= \mathbb{P}_A\{T_C < T\} \mathbb{P}_A\{S_C > T | T_C < T\} \\ &= \mathbb{P}_A\{T_C < T\} \mathbb{P}_A\{T - T_C < J | T > T_C\}. \end{aligned}$$

By the lack of memory property, given $T > T_C$ the law of $\tilde{T} = T - T_C$ is exponential with rate $\lambda = 2$ again. Hence

$$\mathbb{P}_A\{T_C < T, S_C > T\} = \mathbb{P}_A\{T_C < T\} \mathbb{P}_C\{\tilde{T} < J\},$$

where $\tilde{T}$ is an independent exponential random variable with rate $\lambda = 2$. The right hand side can be calculated. We write down

$$\lambda I - Q = \begin{pmatrix} \lambda + 3 & -1 & -2 \\ -3 & \lambda + 3 & 0 \\ -1 & -2 & \lambda + 3 \end{pmatrix},$$

and derive

$$\mathbb{P}_A\{T_C < T\} = \hat{f}_{AC}(\lambda) = \frac{r_{AC}(\lambda)}{r_{CC}(\lambda)} = \frac{\det \begin{pmatrix} -1 & -2 \\ \lambda + 3 & 0 \end{pmatrix}}{\det \begin{pmatrix} \lambda + 3 & -1 \\ -3 & \lambda + 3 \end{pmatrix}} = \frac{2(\lambda + 3)}{\lambda^2 + 6\lambda + 6},$$

and, by Lemma 2.7,

$$\mathbb{P}_C\{J > \tilde{T}\} = \frac{\lambda}{\lambda + 3}$$

recalling that J and $\tilde{T}$ are independent exponentially distributed with rates 3 resp. $\lambda = 2$. Hence,

$$\mathbb{P}_A\{T_C < T, S_C > T\} = \frac{2\lambda}{\lambda^2 + 6\lambda + 6} = \frac{4}{22} = \frac{2}{11}.$$

2.4 Long-term behaviour and invariant distribution

We now look at the long-term behaviour of *continuous* time Markov chains.

Definition 2.18. Let Q be a Q-matrix on the countable state space I . The **jump matrix** of Q is $\Psi = (\psi_{ij} : i, j \in I)$, where

$$\psi_{ij} = \begin{cases} q_{ij}/q_i & \text{if } i \neq j \text{ and } q_i > 0; \\ 0 & \text{if } i = j \text{ and } q_i > 0, \end{cases}$$

and

$$\psi_{ij} = \begin{cases} 1 & \text{if } i = j \text{ and } q_i = 0; \\ 0 & \text{if } i \neq j \text{ and } q_i = 0. \end{cases}$$

Observe that Ψ is a discrete time transition matrix (that describes only the jumps, not the times spent at states).

Definition 2.19. Communicating classes of Q are defined in terms of Ψ . In particular, Q is called **irreducible**, if Ψ is.

Turning to recurrence and transience: we define

$$J_1 := \inf\{t > 0 : X_t \neq X_0\} = \text{'time of first jump'}$$

$$T_i^+ := \inf\{t > J_1 : X_t = i\}.$$

When $X_0 = i$, T_i^+ equals the first return time to i .

Definition 2.20. Let Q be a Q-matrix on I . We call $i \in I$ **recurrent**, if $\mathbb{P}_i[T_i^+ < \infty] = 1$. We call $i \in I$ **transient**, if $\mathbb{P}_i[T_i^+ < \infty] < 1$.

Observe that i is recurrent for Q if and only if i is recurrent for Ψ .

Definition 2.21. When i is recurrent, we call i **positive recurrent**, if $m_i := \mathbf{E}_i[T_i^+] < \infty$. We call i **null recurrent**, if $m_i := \mathbf{E}_i[T_i^+] = \infty$.

2.4.1 The invariant distribution

Recall that in the *discrete* time case the stationary distribution π on I was given as the solution of the equation $\pi P = \pi$ and, by iteration,

$$\pi = \pi P = \pi P^2 = \pi P^3 = \dots = \pi P^n \text{ for all } n \geq 0.$$

Hence we would expect an invariant distribution π of a continuous time Markov chain with transition matrix function $(P(t) : t \geq 0)$ to satisfy

$$\boxed{\pi P(t) = \pi \text{ for all } t \geq 0.}$$

If the statespace is finite, we can differentiate with respect to time to get $\pi P'(t) = 0$. Setting $t = 0$ and recalling $P'(0) = Q$ we get

$$\boxed{\pi Q = 0.}$$

The two boxed statements can be shown to be equivalent in the general case, see Norris Theorem 3.5.5.

Definition 2.22. Let Q be a Q -matrix on the countable state space I . A probability distribution π satisfying $\pi Q = 0$ is called an *invariant distribution*, or sometimes *equilibrium* or *stationary distribution*.

The following technical condition will be necessary: Q is called **non-explosive**, if with probability 1, there cannot be infinitely many jumps in finite time, from any starting state. This will be satisfied in all our examples. Henceforth we assume that Q is non-explosive and irreducible.

Theorem 2.23. *The following are equivalent for an irreducible Q -matrix Q .*

- (i) *every state is positive recurrent;*
- (ii) *some state i is positive recurrent;*
- (iii) *there exists an invariant distribution π and Q is non-explosive.*

When the above hold, we have $\pi_i = \frac{1}{m_i q_i}$, $i \in I$.

Proof. (Non-examinable). See Norris Theorem 3.5.3. □

Theorem 2.24 (Convergence to equilibrium). *Suppose that Q is irreducible and positive recurrent. For any initial distribution w , we have*

$$\lim_{t \rightarrow \infty} \mathbb{P}_w[X_t = j] = \pi_j = \frac{1}{m_j q_j}, \quad j \in I.$$

In particular, $p_{ij}(t) = \mathbb{P}_i[X_t = j] \rightarrow \pi_j$ as $t \rightarrow \infty$ for all $i, j \in I$.

Theorem 2.25 (Ergodic Theorem). *Suppose Q is irreducible and positive recurrent. Let $V_j(t) = \int_0^t \mathbf{1}_{X_s=j} ds$ be the time spent in state j up to time t . We have*

$$\lim_{t \rightarrow \infty} \frac{V_j(t)}{t} = \pi_j \text{ with probability one, for all } j \in I.$$

More generally, if $f : I \rightarrow \mathbb{R}$ is bounded, then with probability 1 we have

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t f(X_s) ds = \sum_{j \in I} f(j) \pi_j.$$

These theorems are analogous to our main theorems in the discrete time case.

Remark 2.26. An important case when the above three theorems always apply is an irreducible Q on a *finite* state space. Then π exists, as a left eigenvector corresponding to the eigenvalue 0 of Q . It can also be shown that on a finite state space there is no explosion¹ Hence the Markov chain is positive recurrent, and we have convergence to π and also the ergodic theorem.

¹(non-examinable) See Norris Theorem 2.7.1.

2.4.2 Symmetrisability

Suppose $m = (m_i : i \in I)$ satisfies $m_i \geq 0$, with m not the zero vector. We say that Q is m -symmetrisable if

$$m_i q_{ij} = m_j q_{ji} \text{ for all } i, j \in I$$

Lemma 2.27. *If Q is m -symmetrizable and $M = \sum_{i \in I} m_i < \infty$, then $\pi_i := m_i/M$ defines an invariant distribution.*

Proof. We have

$$\pi_i q_{ij} = \pi_j q_{ji}, \quad \text{for all } i, j \in I. \quad (*)$$

From this we have

$$(\pi Q)_j = \sum_{i \in I} \pi_i q_{ij} = \sum_{i \in I} \pi_j q_{ji} = \pi_j \sum_{i \in I} q_{ji} = \pi_j \cdot 0 = 0.$$

□

We call (*) the **detailed balance equations**, and say that Q and π are in detailed balance.

Note that, as in the discrete case, the matrix Q may not be symmetrisable, but the invariant distribution may still exist.

Example 2.28. (M/M/1 queue) A single server has a service rate μ , customers arrive individually at a rate λ . Let X_t be the number of customers in the queue (including the customer currently served) and $I = \{0, 1, 2, \dots\}$. The Q -matrix is given by

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & \dots \\ \mu & -\lambda - \mu & \lambda & 0 & \dots \\ 0 & \mu & -\lambda - \mu & \lambda & 0 & \dots \\ \dots & 0 & \ddots & \ddots & \ddots & 0 \end{pmatrix}.$$

(a) We first try to find the invariant distribution using symmetrisability. For this purpose we have to solve the detailed balance equations $m_i q_{ij} = m_j q_{ji}$, explicitly

$$\begin{aligned} m_0 \lambda &= m_1 \mu &\Rightarrow m_1 &= \frac{\lambda}{\mu} m_0, \\ m_1 \lambda &= m_2 \mu &\Rightarrow m_2 &= \frac{\lambda}{\mu} m_1, \\ m_n \lambda &= m_{n+1} \mu &\Rightarrow m_{n+1} &= \frac{\lambda}{\mu} m_n, \end{aligned}$$

for all $n \geq 0$. Hence denoting by $\rho = \lambda/\mu$ the *traffic intensity* we have $m_i = \rho^i m_0$. If $\rho < 1$, then

$$M = \sum_{i=0}^{\infty} m_i = m_0 \sum_{i=0}^{\infty} \rho^i = \frac{m_0}{1-\rho} < \infty,$$

Hence get that

$$\pi_i = \frac{m_i}{M} = (1-\rho)\rho^i \text{ for all } i \geq 0.$$

In other words, if $\rho < 1$ the invariant distribution is the geometric distribution with parameter $1 - \rho$ shifted by -1 .

(b) Over a long time range the mean queue length is

$$\sum_{i=1}^{\infty} i\pi_i = \sum_{i=1}^{\infty} (1 - \rho)i\rho^i = (1 - \rho)\rho \left(\sum_{i=1}^{\infty} \rho^i \right)' = \frac{\rho}{1 - \rho}.$$

PS: In the case $\rho > 1$ the chain is not recurrent, in the case $\rho = 1$ the chain is null recurrent and no invariant distribution exists.

(c) The average amount of time a customer spends in the queue (in equilibrium) can be computed as follows. An arriving customer will find i people in the queue with probability $(1 - \rho)\rho^i$. The time it takes to serve each customer is an $\text{Exp}(\mu)$ random variable, having mean $1/\mu$. Thus the average waiting time is:

$$\begin{aligned} \sum_{i=0}^{\infty} (i+1) \frac{1}{\mu} (1 - \rho)\rho^i &= \frac{1 - \rho}{\mu} \frac{d}{d\rho} \left(\sum_{i=0}^{\infty} \rho^i \right) = \frac{1 - \rho}{\mu} \frac{d}{d\rho} \left(\frac{1}{1 - \rho} \right) = \frac{1 - \rho}{\mu} \frac{1}{(1 - \rho)^2} = \frac{1}{\mu(1 - \rho)} \\ &= \frac{1}{\mu - \lambda}. \end{aligned}$$

(d) A **busy period** is a maximal time period during which the queue is non-empty. The mean length of a busy period can be computed as follows. We have

$$\mathbf{E}_0[T_0^+] = \text{'mean return time to empty queue'} = m_0 = \frac{1}{\pi_0 q_0}.$$

We also have

$$\mathbf{E}_0[J_1] = \text{'mean time queue remains empty'} = \frac{1}{q_0}.$$

Therefore, the mean length of a busy period is given by

$$\mathbf{E}_0[T_0^+] - \mathbf{E}_0[J_1] = \frac{1}{\pi_0 q_0} - \frac{1}{q_0} = \frac{1}{\mu - \lambda}.$$

(e) The average number of customers served in a busy period can be computed as follows. The jump Markov chain is a simple random walk that steps to the right with probability $p = \lambda/(\lambda + \mu)$. We have

$$\begin{aligned} \# \text{ customers served during busy period} &= \# \text{ customers who arrived during the busy period} \\ &= \# \text{ up-steps of the jump Markov chain up to first return to 0} \\ &= \frac{1}{2} T_0^{\text{JMC}}. \end{aligned}$$

From calculation made for simple random walk we have

$$\mathbf{E}_0^{\text{JMC}}[T_0^{\text{JMC}}] = 1 + \mathbf{E}_1^{\text{JMC}}[T_0^{\text{JMC}}] = 1 + \frac{1}{1 - 2p} = \frac{2\mu}{\mu - \lambda}.$$

Thus the average number of customers served is $1/(1 - \rho)$.